. Experiment no:2

Title:-implement sorting merging and slicing operations on list and tuples using python

1. Create a list of 10 integers and sort it in ascending order.
   nums = [8, 3, 9, 1, 5, 2, 10, 7, 4, 6]
   nums.sort() print("Ascending
   order:", nums)

```
Ascending order: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

2. Create a list of 10 integers and sort it in descending order.
   nums = [8, 3, 9, 1, 5, 2, 10, 7, 4, 6] nums.sort(reverse=True)
   print("Descending order:", nums)

```
Descending order: [10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

3.Create a tuple of 6 numbers and sort it in ascending order.

tup = (8, 3, 6, 1, 9, 2) sorted_tup

= tuple(sorted(tup))

print("Sorted tuple:", sorted_tup)

```
Sorted tuple: (1, 2, 3, 6, 8, 9)
```

4.Merge two lists of integers and print the merged list.

list1 = [1, 3, 5, 7] list2 = [2, 4,

6, 8] merged = list1 + list2

print("Merged list:",

merged)

```
Merged list: [1, 3, 5, 7, 2, 4, 6, 8]
```

5. Merge two tuples of numbers and print the resulting tuple.

t1 = (1, 2, 3) t2

= (4, 5, 6)

merged = t1 + t2 print("Merged

tuple:", merged)

```
Merged tuple: (1, 2, 3, 4, 5, 6)
```

6. Take a list of 8 numbers and print the first 4 elements using slicing nums = [11,

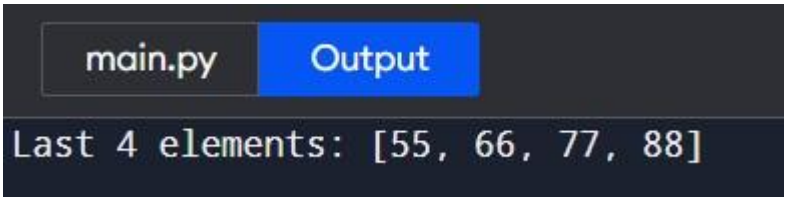
22, 33, 44, 55, 66, 77, 88] print("First 4 elements:", nums[:4])

```
First 4 elements: [11, 22, 33, 44]
```

.7. Take a list of 8 numbers and print the last 4 elements using slicing.

nums = [11, 22, 33, 44, 55, 66, 77, 88]

print("Last 4 elements:", nums[-4:])



8. From a list of 10 elements, print every second element using slicing.

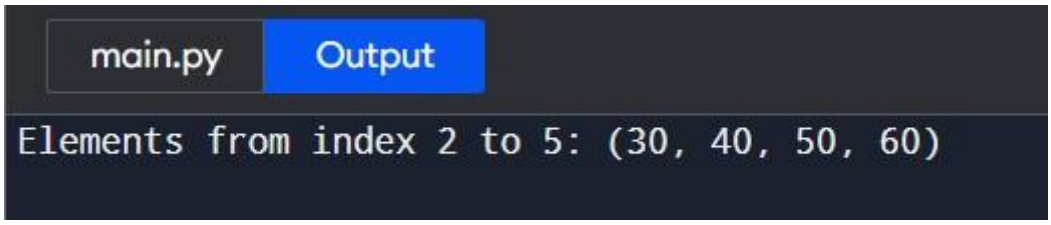
nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

print("Every second element:", nums[::2])



9. From a tuple of 7 elements, print elements from index 2 to 5 using slicing. tup = (10, 20, 30,

40, 50, 60, 70) print("Elements from index 2 to 5:", tup[2:6])



10. From a tuple of 7 elements, print all elements in reverse order using slicing.

tup = (10, 20, 30, 40, 50, 60, 70)

print("Reversed tuple:", tup[::-1])

```
Reversed tuple: (70, 60, 50, 40, 30, 20, 10)
```

11. Combine two lists, sort the combined list in ascending order, and print the result.

a = [4, 1, 8, 3] b

= [6, 2, 5, 7]

combined = a + b combined.sort()

print("Sorted combined list:", combined)

```
main.py  Output
Sorted combined list: [1, 2, 3, 4, 5, 6, 7, 8]
```

12. Combine two tuples, sort the combined tuple in ascending order, and print the result.

t1 = (9, 1, 7) t2 = (6, 4, 2) merged = t1

+ t2 sorted_tuple =

tuple(sorted(merged))

print("Sorted combined tuple:", sorted_tuple)

```
Sorted combined tuple: (1, 2, 4, 6, 7, 9)
```

13. Create a list of strings and sort them alphabetically.

words = ["apple", "banana", "cherry", "date", "fig"] words.sort()

print("Alphabetically sorted list:", words)

```
main.py  Output
Alphabetically sorted list: ['apple', 'banana', 'cherry', 'date', 'fig']

=== Code Execution Successful ===
```

14. Merge a list of strings with another list of strings and print the merged list. 15. Take a list of 10 numbers, split it into two halves, reverse each half separately, list1 = ["red", "blue", "green"] list2 = ["yellow", "pink"] merged = list1 + list2

print("Merged list:", merged)

```
Merged list: ['red', 'blue', 'green', 'yellow', 'pink']
```

15. Take a list of 10 numbers, split it into two halves, reverse each half separately, nums = [1,2,3,4,5,6,7,8,9,10] half = len(nums) // 2

first_half = nums[:half][::-1] second_half = nums[half:][::-1]

print("First half reversed:", first_half)

print("Second half reversed:", second_half)

